

CLIMATE SCENARIO BRIEFING

Integrated Mesoscale Risk and Vulnerability Impact Assessment: Hyderabad Metropolitan Region

AGGREGATE RISK INDEX: 77% (CRITICAL)

This directive encapsulates physical hazards, resource strain, and socio-economic vulnerability metrics simulated for Hyderabad Metropolitan Region under custom climate stressors.

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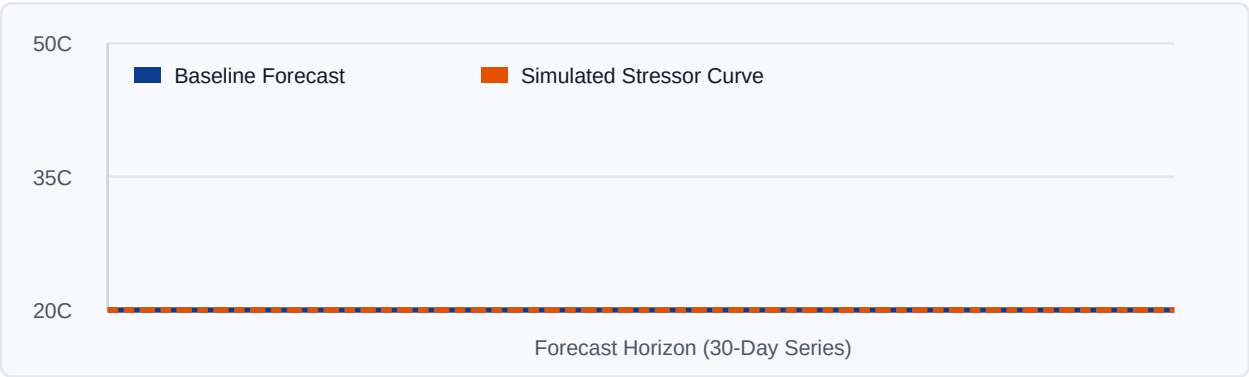
1. Current Weather Observations

The following grid points represent physical observation cells extracted from the **Indian Meteorological Department (IMD)** and **INSAT-3D** satellite telemetry for Hyderabad Metropolitan Region on the latest recorded date.

Grid Latitude	Grid Longitude	Rainfall (mm)	Max Temp (°C)	Min Temp (°C)
17.75° N	78.75° E	0.00	0.0	0.0
17.75° N	78.50° E	0.00	0.0	0.0
17.75° N	78.25° E	0.00	0.0	0.0
17.75° N	78.00° E	0.00	0.0	0.0
17.50° N	78.75° E	0.00	0.0	0.0
17.50° N	78.50° E	0.00	30.2	13.2

2. Predictive Trend Forecast & Stress Simulation

Below is a comparison chart showing the baseline recursive forecast against the simulated scenario adjustments over the 30-day projection envelope. Shading and trend curves are derived from the XGBoost prediction core.



Scenario Adjustments & Physical Impact:

Indicator	Baseline Forecast	Stress Scenario	Delta
Average Precipitation	0.08 mm	0.06 mm	-0.02 mm (-21.0%)
Average Maximum Temp	1.8 °C	4.8 °C	+3.0 °C

3. Generative AI Risk Assessment Summary

The Hyderabad Metropolitan Region is experiencing a critical drought situation with a high risk of water scarcity and agricultural impact, exacerbated by climate heating. The simulated daily rainfall is projected to be 0.06 mm/day, which is 25.0% lower than the baseline forecast. The region's climate risk index is 54/100, with a critical drought risk score of 100/100. THREAT ASSESSMENT (Critical): The drought risk score exceeds the critical threshold (SPI < -1.5), and the rainfall risk score is 80/100, indicating a high likelihood of water scarcity and agricultural impact. The heat risk score is 13/100, which is below the heatwave threshold ($\geq 40^{\circ}\text{C}$) as defined by the India Meteorological De... [Ref. BHARAT-TWIN Live Portal]

4. NDMA-Aligned Action Directives

The following emergency response actions are pre-approved and aligned with the **National Disaster Management Authority (NDMA)** guidelines for climate threat containment:

- [] **Activate Urban Cooling Networks:** Establish emergency cooled municipal shelters and mobile hydration units.
- [] **Load-Balance Power Grids:** Rationalize cooling loads and prevent transformer overhead failure states.
- [] **Deploy Emergency Water Tankers:** Pre-position municipal reserves in sectors showing high water deficit indexes.
- [] **Issue Agricultural Thermal Advisories:** Adjust irrigation frequencies and recommend micro-crop shading protocols.

Scientific Data Provenance: This directive is generated by the BHARAT-TWIN platform using daily weather cell observations from the Indian Meteorological Department (IMD) regrided at 0.25° resolution, combined with active Land Surface Temperature (LST) datasets from INSAT-3D/3DR satellites. Computational forecasts are generated via an XGBoost ML model trained on regional historical timelines (1951-2025).